Precambrian Research, 28 (1985) 75—110
Elsevier Science Publishers B.V., Amsterdam — Printed in The Netherlands

PALAEOMAGNETIC STUDY OF THE NAGSSUGTOQIDIAN MOBILE BELT IN CENTRAL—WEST GREENLAND

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(Received June 6, 1983; revision accepted August 20, 1984)

ABSTRACT

Piper, J.D.A., 1985. Palaeomagnetic study of the Nagssugtoqidian mobile belt in central—West Greenland. Precambrian Res., 28: 75—110.

This paper reports a palaeomagnetic study of amphibolite and granulite facies basement rocks across a 60 km wide north—south traverse through the Nagssugtoqidian mobile belt in West Greenland, which incorporates the c. 2500 Ma Itivdleq shear belt, the c. 1800 Ma Ikertôq shear belt, and adjoining terrains. It also integrates the results with earlier investigations of the belt to produce a composite record of the magnetic field during uplift and cooling of the terrain through c. 600—500°C at c. 1750—1600 Ma. Progressive a.f. and thermal cleaning of samples from 81 sites identifies a range of magnetic components. Lowest blocking temperature remanences in Nagssugtoqidian (c. 1700 Ma) amphibolite facies rocks yield steep W to N positive components residing in magnetite. Granulite facies rocks formed at c. 1800 Ma generally have a dual record incorporating steep N to NW positive components unblocked by 580°C and shallow E—W directed components of dual polarity only unblocked close to the Curie point of haematite. A common feature of relict c. 2500 Ma amphibolite and granulite facies terrains is a progressive movement of sample directions back along a palaeofield migration path with treatment and is interpreted to represent the recovery of a discrete spectrum acquired during a.p.w. Areas with a distributed spectra through the critical range of blocking temperatures, altitude-controlled sections, and blocking temperature relationships all record consistent a.p.w. during the interval of uplift-related cooling. The collective data define a clockwise palaeofield migration path between D = 174, I = 32°, and D = 348, I = 66° referred to as the 'A' magnetisation. The equivalent palaeopoles define one limb of an a.p.w. loop conforming to other Laurentian poles assigned to the interval 1750—1650 Ma and widely represented as a post-Hudsonian overprinting or cooling remanence. Shallow E—W directions, designated 'B', predominate in late tectonic diorites and are therefore younger than 1700 Ma; they appear to represent a localised reheating event near the northern margin of the Ikertôq shear belt, possibly related to this magmatism, which produced the hemoilmenite phases. The equivalent palaeopoles are provisionally interpreted to represent part of the return path of the APW loop because they are coincident with other poles dated c. 1630 Ma. Collectively the palaeofield migration data from all studies define a thermal dome broadly coincident with the Ikertôq shear belt, but it is not possible to isolate the contribution of components of vertical movement along faults running through the belt to this regional effect.

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INTRODUCTION

The Nagssugtoqidian mobile belt, a zone of Archaean gneisses reworked during Proterozoic times, is exposed in both East and West Greenland, and is presumed to have an arcuate form around the northern margin of the South Greenland Archaean nucleus (Bridgwater et al., 1973). In West Greenland it is both well exposed and accessible and has been the subject of palaeomagnetic endeavour by a number of workers, firstly to investigate the effect of slow uplift-related cooling on the acquisition of magnetic remanence in plutonic rocks (Morgan, 1976a, b; Beckmann et al., 1977; Piper, 1981), and secondly to elucidate the apparent polar wander (a.p.w.) record during the lower–middle Proterozoic consolidation of this terrain (Beckmann and Mitchell, 1976; Fahrig and Bridgwater, 1976; Beckmann, 1979). The present study reports results of a detailed regional palaeomagnetic sample across the Nagssugtoqidian belt in West Greenland with the broad objectives of (1) resolving the magnetic record from the different metamorphic facies and tectonic zones, (2) linking together into a time sequence with the separate magnetisations identified from this and earlier studies, and (3) bringing together collective results from the Nagssugtoqidian terrain in Greenland to analyse them in the context of the geochronologic data and the regional palaeomagnetic record from the Laurentian Shield as a whole.

OUTLINE OF GEOLOGY AND GEOCHRONOLOGY

In common with the major cratonic nucleus immediately to the south (Bridgwater et al., 1973), the rocks within the mobile zone are Archaean in origin and stabilised in granulite and amphibolite facies assemblages in Archaean times (Korstgaard, 1979). Transcurrent dextral shearing, previously referred to as the Nag 1 episode, later took place within spaced subvertical ENE zones of intense shearing to impose a strong tectonic fabric within a zone ~ 40 km wide. It is bounded on both sides by largely unaffected granulite-facies rocks (although this near-coincidence of deformational and metamorphic boundaries is not found in the Nordre Strømfjord belt 120 km to the north), and it incorporates a 12 km wide augen-like block at Kingâq (Fig. 1) where relict granulite facies rocks are effectively undeformed. Structural and geochronological work (see below) draw an important distinction between the Itivdleq belt in the south and the Ikertôq belt in the north, although the latter appears to be partly superimposed on the former at its easterly extension (Grocott, 1979, fig. 4).

The regional change in orientation of a dyke swarm referred to as the Kangamiut dykes (dated 1950 ± 60 Ma, Kaalsbeek et al., 1978) in the vicinity of the Itivdleq belt was originally thought to indicate progressive deformation towards the north and was used to map the southern boundary of this belt (Noe Nygaard and Ramberg, 1961) but Watterson (1974) found

that the Kangamiut dykes north of the Archaean craton were intruded with an ENE strike controlled by a pre-dyke tectonic fabric; igneous textures show that the dykes have not been reoriented. It is now accepted that the NNE and ENE dykes are components of a conjugate swarm (Escher et al.,

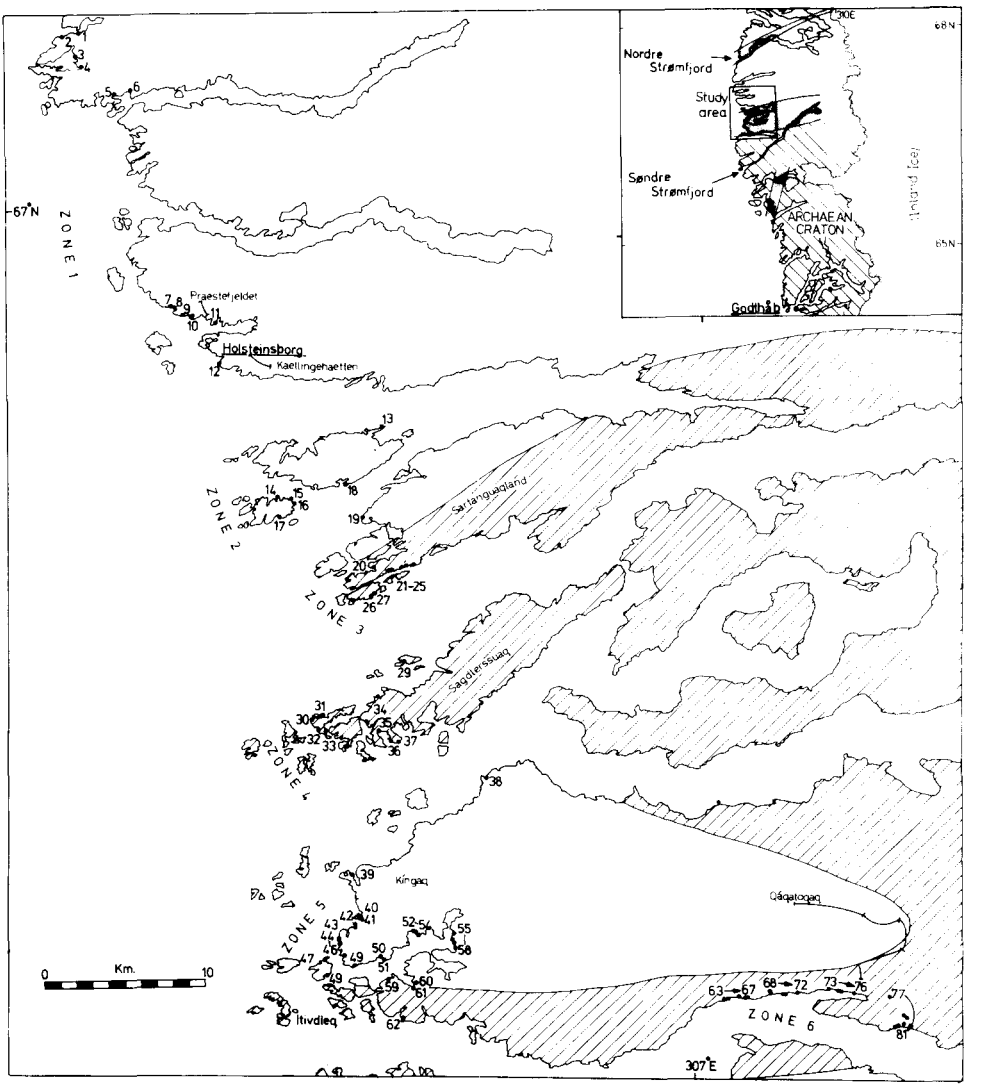


Fig. 1. Location map of the palaeomagnetic sampling sites within the Nagssugtoqidian terrain of West Greenland. The shaded area shows the distribution of amphibolite facies terrain after Bak et al. (1976); the blank area is granulite facies. The dashed lines are tectonic boundaries after Grocott (1979); note that these conform in part to the metamorphic facies boundaries. The inset map shows the regional location of the study area in West Greenland. The major zones of late Archaean and early Proterozoic ductile shear are stippled and the area of oblique shading is the Archaean craton of South Greenland.

1976). In the north, the non-consistency of cross-cutting relationships suggests that they are essentially of the same age, although in the south the NNE dykes consistently cut the E—W dykes and they may be appreciably younger in age (Escher, et al., 1976). Although dykes exhibit chilled margins, intrusion evidently took place into country rock at elevated temperatures because the dykes developed primary pinch and swell structures in the more ductile amphibolites which contrast with the more characteristic parallel-sided dykes in the less-ductile granulite facies terrain (Watterson, 1974; Nash, 1979); it is this ductility contrast which accounts for the regional south to north transitional form of the dykes. Much of the previous palaeomagnetic sampling has been concentrated on these dykes (Morgan, 1976a; Beckmann and Mitchell, 1976; Fahrig and Bridgwater, 1976).

A more localised tectonic episode, originally referred to as "Nag 2", deformed the Kangamiut dykes north of the fjord Kangerdluassuak at 66° 43′ N. It produced intense ductile overthrusting in a SSE direction and incorporated belts of low amphibolite—greenschist facies supracrustal rocks. It refolded Nag 1 structures, and reconstituted the metamorphic assemblages. North of Holsteinsborg, diorites were intensively intruded as dykes and other minor bodies during the last vestiges of ductile deformation associated with this episode; analysis of the form of these intrusions relates the causative stress field to a simple shear (Davidson and Park, 1978).

The most recent geochronological work has clarified an important distinction between two episodes of shear belt development in West Greenland. The Nagssugtoqidian *sensu stricto* is now confined to tectonic reworking in the Ikertôq belt and the entire zone north of Holsteinsborg including the Nordre Strømfjord belt where Rb—Sr systems were reset at 1800 Ma; this is essentially the former "Nag 2" division. The current best estimate for the age of the Nagssugtoqidian and the accompanying granulite facies metamorphism is a Rb—Sr isochron of 1840 ± 22 Ma on syntectonic granite layers in the Nordre Strømfjord Shear zone (M.H. Hickman, personal communication, 1983). Hickman (1979) also provisionally dated a syn-Nagssugtoqidian pegmatite at 1670 ± 150 Ma and late Nagssugtoqidian diorites at c. 1700 Ma by Rb—Sr determinations. These ages are likely to define the maximum age of the magnetisations because the granulite assemblages in the vicinity of Holsteinsborg indicate temperature and pressure conditions of 900—800°C and $\geqslant$ 9.5 kbar (Davidson, 1979). The ages do not, however, bracket the thermal history of the coastal regions very closely because they come from inland locations where the metamorphic conditions were generally less extreme. However, whole rock Rb—Sr ages of 1740 ± 125 and 1600 ± 90 to 1690 ± 10 Ma in feldspars from late to post-tectonic pegmatites in the Nordre Strømfjord area probably represent cooling ages and temperatures of 450—500°C (M.H. Hickman, personal communication, 1983; Hart, 1981) and are thus closely relevant to the magnetisation history. Since the diorites are late tectonic but pre-date regional cooling, the best current estimate of their age is 1750 ± 100 Ma and the estimates quoted above will require further work for precise definition.

In 1979, Hickman reported a preliminary Rb—Sr determination of 2500 ± 40 Ma on a pegmatite formerly referred to as the Nag 1 episode, and further work on the gneisses of the Itivdleq area has given Rb—Sr whole rock ages of around 2500 Ma with little sign of later disturbance (M.H. Hickman, personal communication, 1983). Thus, the Itivdleq shear belt in the south appears to be substantially older than the Ikertôq belt in the north (Fig. 1).

The history of post-tectonic uplift is not yet well understood, but in common with the Lewisian terrain of NW Scotland (e.g., Dickinson and Watson, 1976) it was fairly rapid: K—Ar biotite ages of 1790—1650 Ma from the Nagssugtoqidian belt as a whole (Larson and Møller, 1968) are indicative of closure temperatures of 300—200°C and, comparable to the Rb—Sr dates indicative of temperatures of 500—450°C. Beckmann and Mitchell (1976) determined K—Ar isochron plots of 1620 ± 50 Ma (mean of concordant ages between 1660 and 1580 Ma) from Sagdlerssuaq and 1700 ± 40 Ma and 1830 ± 20 Ma from Itivdleq; these latter ages are again most appropriately interpreted as cooling ages and the authors suggest a difference in cooling age of the order of 100 Ma between the two regions. A K—Ar plot of 2410 ± 40 Ma from the terrain south of Itivdleq Fjord (Beckmann and Mitchell, 1976) suggests that regional cooling may have taken place here considerably before the terrain north of the fjord, and it remains a possibility that some of the magnetisations identified by Fahrig and Bridgwater (1976) from the Archaean craton were of lower Proterozoic age in accordance with their interpretation.

PALAEOMAGNETIC RESULTS

Methodology

A palaeomagnetic site usually comprises a minimum of seven cores distributed over several metres of outcrop. They were cored with a portable drilling motor of the kind described by Doell and Cox (1967) and oriented *in situ* by sights on the sun, in few instances where this was not possible, orientation was by magnetic compass in conjunction with topographic sights. Samples were measured in the laboratory with an automated parastatic magnetometer. Alternating field (a.f.) demagnetisation by tumbling was utilised as the standard method for the treatment because it proved most appropriate for separating the rather broad range of coercivities present in most cores. Samples were progressively demagnetised in steps of 5 or 10 mT (milliTesla) to peak fields of up to 140 mT. Progressive thermal demagnetisation was conducted on separate slices of a palaeomagnetic core from each site in steps of 100 to 500°C and then in steps of 30 to 620 or 680°C; this thermal demagnetisation was followed up in other cores when it identified components not separated by the a.f. method. Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature) were made on small samples from each site as an aid to identification of magnetic phases and the results are summarised in the data tables.

In general, any component with a coercivity of < 15 mT is considered likely to be of secondary viscous origin and is not discussed further here. However, different components are sometimes resolved at single sites over differing ranges of the coercive force and/or blocking temperature spectra. These are best resolved by a combined study of vector and directional plots but cannot always be effectively separated by cleaning techniques notably when both components have similar high blocking temperature spectra. Group mean directions have been calculated (see Table I) when components are considered to be adequately separated such that the end point can be considered to be within the 95% confidence circle and follows the method of Dagley and Ade-Hall (1970). An important feature identified by Morgan (1976b) in some rocks of this area is the consistent and progressive migration of sample mean directions along a small circle. Morgan interpreted this in terms of the gradual subtraction of lower blocking temperature components by the cleaning technique such that the sample direction recovers progressively older parts of the a.p.w. path. This migration is observed in the same sense with altitude in certain vertical sections which appear to confirm Morgan's interpretation (Piper, 1981). The vector plots for such samples show the vectors migrating in towards the origin along a broad arc rather than about an average straight line as they would do if only a undirectional component was being subtracted. Both stereoplots and vector plots are utilised to display sample behaviour here (see Figs. 2—4). Theoretical aspects of the acquisition of magnetic remanence under conditions of slow uplift-related cooling have been discussed by a number of workers (Ueno et al., 1975; Morgan, 1976b; Pullaiah et al., 1975). More refined assessments which take into account the decrease in activation energy of magnetic grains towards the Curie point show that the range of critical blocking temperatures is unlikely to be depressed under slow cooling far below the range determined in the laboratory (Dodson and McClelland-Brown, 1980). Practical studies of slowly-cooled rocks (e.g., Morgan, 1976b; Buchan and Dunlop, 1976; Piper, 1981) have emphasised the most characteristic feature of palaeomagnetism in plutonic terrains, namely the resolution of a sequential migration of palaeofield directions in place of the single coherent directions isolated from rapidly-cooled situations. This variation is variously identified within single cores, between cores from the same site with different coercivity or blocking temperature spectra, and between regional distributions of sites, and it is clear from large areal assessments of palaeomagnetism in such Proterozoic plutonic terrains as the Grenville (McWilliams and Dunlop, 1978; Roy and Blythe-Robertson, 1979), the Sveconorwegian (Stearn and Piper, 1984) and the Lewisian (Smith and Piper, 1982).

In principle the acquisition of magnetisations in slowly-cooled terrains should have been sufficiently slow to average out the effects of secular variation, and at any given crustal level samples with higher blocking temperatures should record earlier parts of the a.p.w. path from those with

lower blocking temperatures. In addition, if the blocking temperature spectrum is well distributed, demagnetisation should recover progressively higher blocking temperature, and therefore older, segments of the total magnetisation so that sample directions move progressively back along the a.p.w. path with thermal cleaning. Although the coercivity spectrum cannot of course, be directly linked to the blocking temperature spectrum, similar behaviours are observed with a.f. and thermal techniques in samples with broad spectra (Morgan, 1976a, b; Piper, 1981) and the phenomenon can, at least in some rocks, be recovered by both methods; indeed, the a.f. method is frequently found to be more discriminating because movements resolved over a whole range of a.f. fields are frequently defined over a narrow range of high temperature demagnetisation. The identification of these effects requires that the components are recognised to have a singular and primary origin.

Summary of previous palaeomagnetic results

The study by Morgan (1976a, b) analysed the detailed palaeomagnetic behaviour of four gneiss and eighteen dyke sites in the vicinity of Itivdleq Fjord. The collective mean direction is D = 207, I = 54 (k = 46) although this group mean calculation does not accommodate the continuous directional changes observed during progressive cleaning. Site mean palaeopoles lie progressively further back along a NW—SE a.p.w. track going from SSE to NNW across Itivdleq Fjord suggesting a regional tilt to the NNW. Three sites distributed over 2 km at 53° 29'W, 66° 38'N which did not conform with this trend, suggested upfaulting of this part of the Kingâq augen. Beckmann and Mitchell (1976) listed further data from the Kangamiut dykes at Itivdleq, together with results from basic gneiss bands and Kangamiut dykes in Sagdlerssuaq and adjoining islands with generally lower coercivities and blocking temperature spectra; nine sites yield a mean of D = 253, I = 81 (k = 52) and confirmed a result of Fahrig and Bridgwater (1976) from 3 sites here.

Beckmann et al. (1977) conducted a further palaeomagnetic study of gneisses up a vertical section of Mt. Kaellinghaetten near Holsteinsborg, and along a horizontal section in Nordre Strømfjord. The mean palaeomagnetic directions determined from the units with lower anisotropy in these two areas yielded mean directions of D = 209, I = 71 and D = 241, I = 57, respectively. In a further investigation of the altitude dependence of magnetic remanence in these terrains, Piper (1981) utilised rocks intruded at late- to post-tectonic phases with minimal fabric and recognised a variation with altitude dependent on blocking temperature spectra. Amphibolite facies terrain at Qaqatoqâq near the eastern margin of the Kingâq augen illustrated systematic migration of sample directions with treatment and a general distribution with altitude which conformed with this trend when subdivided according to blocking temperature spectra. Diorites from

north of Holsteinsborg had generally higher and narrower blocking temperature spectra which also showed a systematic change with altitude. Overall group mean directions from three areas ranged through an arc from D = 168, I = 41 to D = 304, I = 77, with a distinct high blocking temperature component of shallow inclination (D = 77, I = −27) and dual polarity recognised at some sites.

Little palaeomagnetic information is currently available from the less accessible Nagssugtoqidian terrain in East Greenland. Beckmann (1983) gave palaeopoles from gneisses at Angmagssalik and Nordfjord at 274° E, 25° N and 247° E, 14° S. These poles are linked to an estimated age of 1800–1600 Ma for the Angmagssalik Complex and an older, but probably still Nagssugtoqidian, age for the Nordfjord rocks. A general feature of the palaeomagnetic studies in the Nagssugtoqidian terrain has been the recognition of similar directions of magnetisation from late- to post-tectonic intrusions and the country rock gneisses (Morgan, 1976b; Beckmann and Mitchell, 1976; Fahrig and Bridgwater, 1976; Beckmann, 1977), implying that they are uplift-related in both divisions. However, this result requires that both rock types have similar blocking temperature spectra and exceptions have been noted when this is not the case (Piper, 1981).

Areal distribution of the palaeomagnetic collection

Since the structural history, the rock types available for study, and probably the history of uplift, vary from area to area across the Nagssugtoqidian belt, it is subdivided here into six sections (Fig. 1). Zone 1 (sites 1–12) incorporates the granulite facies terrain of Nagssugtoqidian age and sampling was concentrated in diorite bodies intruded in the late ductile stage of the deformation. Zone 2 (sites 13–19) is in granulite facies terrain along the northern margin of the Ikertôq shear belt; sampling here was concentrated mostly in basic and ultrabasic lenses in the regional gneisses. Zone 3 (sites 20–28) is in amphibolite facies terrain on Sarfanguaqland along the northern margin of the Ikertôq belt. Zone 4 is an amphibolite facies terrain of Nagssugtoqidian age near the axis of the Ikertôq belt on Sagdlerssuaq and adjoining islands; highly deformed acid and basic gneiss here incorporate belts of supracrustals. Zone 5 includes two sites (39 and 40) in gneisses and eighteen sites in dykes (41–58) within the Kingâq augen; the granulite assemblage here is considered to be of Archaean age (Korstgaard, 1979). Sites 59–62 continue this section southwards into the Itivdleq shear zone. Zone 6 (sites 63–81) is a continuation of this belt further eastwards along Itivdleq Fjord; the amphibolite facies assemblage in the country rocks here is considered to be of Archaean (Nag 1) age (Korstgaard, 1979).

Zone 1, North of Amerdlôq

Diorite dykes 1, 4 and 6 have steep components illustrating an exponential fall with a.f. demagnetisation but moderately high blocking temperatures with an appreciable component remaining above the Curie point of magnetite. Site 2 exhibits more complex behaviour: a coherent direction in four samples (i) is smeared in other samples along a small circle towards a direction similar to site 6, a dispersal not appreciably reduced by thermal demagnetisation. At site 3 directions migrate progressively with demagnetisation from (i) to (ii); three further samples migrate to a shallow westerly component (iii) stable to thermal cleaning above 600° C. The site mean directions from these six diorites lie near the younger part of the directional trend defined by an altitude section up the mountain Igdlunguit qulât 10 km to the NE (Piper, 1981) and sites 2 and 3 illustrate directional migrations consistent with this trend, and their lower altitudinal position.

Sites 7—12 in late tectonic diorites around the two of Holsteinsborg have a remanence not appreciably influenced by a.f. treatment (Fig. 3) and dominated by high blocking temperature components residing in hemoilmenite; the exception is site 11 where four low-coercivity samples have a remanence (i) eliminated near the Curie point of magnetite. Most cores also have a steep inclination low-coercivity component of probable Mesozoic to Recent origin which is progressively subtracted by cleaning (Figs. 2 and 3); other samples exhibit only the single shallow component and retain this to the Curie point of haematite. Thus the sites around Holsteinsborg and the high blocking temperature component at site 3 define a shallow ENE—WSE axis of dual polarity which is similar (but with more positive inclination) to the dual polarity axis defined by sites up the mountain Praestfjeldet opposite Holsteinsborg. This is probably a regional mineralogical effect within this area because rocks to the north (sites 1—6) and south (Kaellinghaeten, Beckmann et al. (1977) and Zone 2) have lower blocking temperature steep components lying near the trend resolved from the altitude sections.

Zone 2, northern border to Ikertôq shear belt

These granulite facies rocks from the Manitsorssuaq, Umanârssugssuaq and the northern part of Sarfanguaqualand show a more complex behaviour to demagnetisation compared with rocks to the north and south, an observation which may be related to their structural position at the northern border of the Ikertôq shear belt. At site 13 (i) is a coherent component progressively subtracted by a.f. treatment; a high blocking temperature component (ii) is recovered at high stages of thermal treatment in three cores. Sites 14, 17 and 19 have similar components removed by thermal treatment close to the Curie point of magnetite; the complexities are a low blocking temperature component (i) removed below 400° C at site 15

TABLE I

Palaeomagnetic results from the Nagssugtoqidian terrain in central—West Greenland

Site No.		N	R	D	I	k	Palaeomagnetic Pole °E	Palaeomagnetic Pole °N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
(i) Zone 1, Metamorphic facies North of Ikertôq Shear Belt, North of Amerdlôq											
1	G	8	7.98	268	72	325	249	50	5/5	1.899 - 0.671	560, 680
2	G(i)	5	4.98	242	62	182	259	30	7/9	2.348 -05.67	570
	(ii)	4	3.99	199	68	585	293	28	5/6		
3	G(i)	5	4.74	225	55	15	269	18	20/28	3.161 - 0.396	570, 680
	(ii)	5	4.79	216	51	19	276	13	17/25		
	(iii)	3	2.98	250	12	100	237	-2	6/13	2.084 - 1.021	560
4	G	8	7.92	322	80	85	247	76	11/12	2.084 - 1.021	560
5	G	8	7.95	278	68	151	237	49	6/8	1.543 - 0.473	560
6	G	7	6.98	181	76	388	306	40	5/6	1.253 - 0.394	570
7	G	6	5.97	76	6	179	48	8	3/5	1.324 - 0.655	570
8	G	7	6.98	77	7	287	47	8	2/4	2.643 - 1.035	570, 680
9	G	7	6.90	62	5	604	62	13	4/8	2.118 - 1.353	560, 640
10	G	9	8.93	75	2	121	50	7	2/5	3.075 - 2.491	560, 650
11	G	4	3.98	301	59	127	207	48	9/12	2.144 - 0.565	560
12	G	7	6.94	80	25	99	41	16	4/7	0.829 - 0.058	570

Site No.		N	R	D	I	k	Palaeomagnetic Pole °E	Palaeomagnetic Pole °N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
(ii) Zone 2, Ikertôq Shear Belt, northern margin, granulite facies											
13	G(i)	8	7.92	279	74	86	249	53	10/11	57.391 - 0.084	
	(ii)	3	2.99	253	10	214	234	-2	4/10		
14	G	5	4.98	313	79	184	241	72	10/11	0.360 - 0.096	570
15	G(i)	5	4.83	2	3	24	124	25	8/16	0.072 - 0.033	570, 670
	(ii)	5	4.84	277	57	25	228	37	17/23		
16	G(i)	6	5.98	325	76	245	223	75	7/8	1.243 - 0.126	580
	(ii)	2	1.99	148	49	2033	334	10	5/7		
17	G	8	7.84	304	72	43	223	63	13/15	1.230 - 0.792	550
18	G(i)	4	3.96	305	-8	76	182	9	5/11	4.295 - 0.019	565, 670
	(ii)	2	1.99	51	-47	2986	82	-12	4/6		
19	G(i)	6	5.80	300	62	25	210	51	16/21	2.479 - 0.002	570, 640
	(ii)	4	3.92	86	46	37	28	27	12/19		
(iii) Zone 3, Ikertôq Shear Belt, Sarfanguaqland											
20	A(i)	3	2.99	358	60	413	130	64	7/9	5.242 - 0.118	570, 640
	(ii)	4	3.98	83	6	130	42	6	4/8		
21	A	7	6.97	344	63	176	155	65	6/7	0.937 - 0.207	565
22	A	7	6.97	325	68	202	190	67	6/7	1.509 - 0.214	
23	A(i)	6	5.98	289	60	251	219	45	5/6	3.799 - 0.995	560, 660
	(ii)	4	3.99	207	56	414	284	15	5/7		

TABLE I (*continued*)

Site No.		N	R	D	I	k	Palaeomagnetic Pole °E	Palaeomagnetic Pole °N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
24	A	7	6.99	295	62	514	215	49	3/4	5.394 - 3.574	570
25	A	7	6.96	288	63	161	222	47	6/8	3.645 - 1.528	570
26	A(i)	6	5.93	350	67	70	149	72	11/13	2.473 - 1.241	570, 680
	(ii)	5	4.73	85	-17	15	42	-7	11/22		
27	A(i)	6	5.93	321	63	69	190	60	10/13	2.704 - 1.892	565, 660
	(ii)	8	7.71	46	-1	24	79	15	6/12		
28	A	8	7.96	282	58	171	24	40	5/6	4.478 - 2.771	570
(iv) Zone 4, Sagdlerssuaq											
29	A	6	5.90	271	68	52	243	46	13/16	0.124 - 0.010	550
30	A(i)	8	7.99	280	73	578	243	56	4/4	1.762 - 0.055	555
	(ii)	4	3.98	216	71	96	282	35	14/16		
31	A(i)	7	6.81	242	86	31	292	62	22/22	2.860 - 1.114	555
	(ii)	7	6.14	83	19	7	39	12	13/26		
32	A(i)	6	5.32	303	46	7	197	38	22/34	0.042 - 0.010	320, ——
	(ii)	4	3.97	216	71	96	282	35	14/16		
33	A(i)	3	3.57	262	60	15	242	34	38/50	0.082 - 0.010	560, 670
	(ii)	6	5.74	92	25	19	30	11	9/17		
34	A(i)	7	6.91	275	74	65	240	55	12/14	0.020 - 0.012	550
	(ii)	4	3.92	273	54	39	229	32	16/21		
35	A	8	7.99	267	73	601	252	51	4/43	0.561 - 0.051	570

Site No.		N	R	D	I	k	Palaeomagnetic Pole °E	Palaeomagnetic Pole °N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
36	A	9	8.81	113	73	43	349	45	13/14	0.397 - 0.015	560
37	A	8	7.87	232	67	54	269	32	10/13	1.263 - 0.444	570
38	G(i)	7	6.79	265	72	29	250	48	18/20	32.425 - 0.043	530
	(ii)	7	6.78	249	63	25	255	33	15/19		
(v) Zone 5, Itivdleq Shear Belt, Itivdleq district											
39	G	8	7.56	175	58	16	311	16	16/21	2.014 - 9.511	570
40	G(i)	7	6.98	187	61	349	301	19	4/5	2.128 - 0.266	540
	(ii)	7	6.99	184	54	3100	303	11	1/2		
41	G(i)	7	6.92	214	56	84	279	17	7/10	4.635 - 0.939	550
	(ii)	7	6.93	192	51	82	296	9	6/9		
42	G(i)	6	5.98	229	79	289	282	49	7/8	1.478 - 0.956	560
	(ii)	6	5.99	204	71	517	290	33	4/5		
43	G(i)	6	5.98	181	36	279	306	-3	3/5	0.389 - 0.169	565
	(ii)	7	6.99	181	31	934	306	-7	1/2		
44	G(i)	8	7.99	190	60	507	300	10	2/3	2.626 - 0.736	570
	(ii)	8	7.96	187	52	163	299	17	5/7		
45	G(i)	8	7.95	188	47	147	302	1	4/6	0.748 - 0.386	570
	(ii)	8	7.97	186	42	214	299	5	3/5		
46	G(i)	6	5.97	197	60	160	307	13	5/8	9.108 - 1.424	570
	(ii)	6	5.89	180	55	46	293	18	12/15		

TABLE I (*continued*)

Site No.		N	R	D	I	k	Palaeomagnetic Pole °E	Palaeomagnetic Pole °N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
47	G	9	8.99	198	64	813	293	23	2/3	6.559 - 2.837	570
48	G	6	5.99	165	57	959	319	15	2/3	1.918 - 0.243	565
49	G(i)	3	2.99	254	64	482	251	35	7/9	9.265 - 4.393	570
	(ii)	4	3.97	194	74	96	298	37	15/17		
50	G	6	5.95	243	57	105	284	19	7/10	5.368 - 6.341	560
51	G(i)	4	3.99	205	56	634	298	8	3/5	11.567 - 1.484	560
	(ii)	7	6.99	191	50	406	285	15	3/4		
52	G(i)	4	3.99	224	63	714	274	26	4/5	7.199 - 2.042	560
	(ii)	4	3.99	195	65	206	295	25	8/10		
53	G(i)	4	3.99	195	65	182	302	30	10/12		
	(ii)	5	4.98	187	70	199	295	24	7/9		
54	G(i)	8	7.99	229	61	1603	267	25	2/2	7.127 - 4.705	550
	(ii)	7	6.99	212	61	1267	281	21	2/3		
55	G(i)	8	7.93	217	77	99	286	45	10/11	2.822 - 0.506	540
	(ii)	4	3.83	228	37	18	263	4	15/26		
56	G(i)	8	7.98	174	57	281	312	15	4/5	3.096 - 1.270	540
	(ii)	7	6.99	180	57	400	307	14	3/4		
57	G(i)	6	5.98	235	65	202	265	31	6/8	3.182 - 1.171	530
	(ii)	7	6.91	244	59	67	255	27	8/11		

Site No.		N	R	D	I	k	°E	°N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
58	G(i)	7	6.97	200	64	180	292	23	6/7	3.580 - 1.643	530
	(ii)	8	7.98	193	58	343	297	16	3/4		
59	G(i)	7	6.96	209	51	147	282	11	4/7	1.934 - 1.206	565
	(ii)	7	6.99	202	45	1737	287	5	1/2		
60	G(i)	7	6.98	159	64	280	323	23	5/6	6.712 - 4.286	565
	(ii)	7	6.98	168	58	361	317	15	3/5		
61	G(i)	8	7.97	176	62	215	309	20	5/6	2.758 - 1.729	560
	(ii)	9	8.97	175	58	308	311	15	3/4		
62	A(i)	6	5.99	198	64	344	294	30	5/6	11.811 - 7.757	570
	(ii)	6	5.99	190	68	520	299	28	4/5		
(vi) Zone 6, Itivdleq Shear Belt, Itivdleq Fjord											
63	A	8	7.99	231	62	935	269	27	2/2	19.236 -11.450	570
64	A	–	—	—	—	—	—	—	—		
65	A	7	6.97	178	62	196	310	20	5/7	0.020 - 0.010	580
66	A(i)	7	6.55	192	78	14	300	43	29/32	0.676 - 0.299	585
	(ii)	8	7.72	170	68	25	314	28	16/19		
67	A(i)	8	7.98	264	69	384	249	45	4/5	1.214 - 0.473	575
	(ii)	7	6.97	205	66	188	289	26	6/7		
68	A(i)	7	6.91	258	55	64	243	28	8/11	0.096 - 0.011	580
	(ii)	5	4.55	249	44	9	246	16	21/34		

TABLE I (*continued*)

Site No.		N	R	D	I	k	°E	°N	dp/dm	J_o ($\times 10^{-5}$ A.m^2 kg^{-1})	Curie Point (°C ± 10°)
69	A(i)	8	7.91	220	55	82	274	17	6/9	5.156 - 0.617	575
	(ii)	8	7.89	217	52	67	276	13	6/9		
70	A	7	6.91	232	58	64	265	23	8/11	1.117 - 0.325	575
71	A	7	6.98	288	71	354	234	56	5/6	0.377 - 0.106	580
72	A(i)	7	6.90	239	59	59	260	25	9/12	16.430 - 2.992	575
	(ii)	7	6.88	222	59	50	274	21	10/13		
73	A	5	4.95	53	79	73	7	71	16/17	0.040 - 0.026	580
74	A	5	4.76	334	74	17	202	77	31/34	4.605 - 0.033	570
75	A(i)	6	5.97	235	77	145	277	48	10/10	0.741 - 0.034	580
	(ii)	6	5.93	220	76	70	284	43	14/15		
76	A(i)	4	3.96	277	52	72	225	32	10/15	0.016 - 0.004	575
	(ii)	3	2.98	203	61	101	290	20	14/19		
77	A(i)	7	6.96	224	65	161	274	29	6/8	4.000 - 2.082	565
	(ii)	3	2.92	175	24	25	312	-11	14/27		
78	A	7	6.96	224	65	161	274	29	6/8	0.024 - 0.001	560
79	A	8	7.88	199	60	60	292	17	8/11	1.303 - 0.062	580
80	A	—	—	—	—	—	—	—	—	1.497 - 0.313	560
81	A(i)	7	6.93	229	67	85	272	32	9/11	1.388 - 0.240	575
	(ii)	6	5.97	203	58	163	287	17	6/8		

D is the mean declination and I is the mean inclination of the magnetic vector derived from the mean of N samples yielding a resultant vector R. k is the precision parameter (=(N-1)/N-R). Dp and dm are the semi axes of the oval of 95 percent confidence about the derived paleomagnetic pole in the colatitude direction and at right angles to its respectively. G = Granulite facies. A = Amphibolite facies.

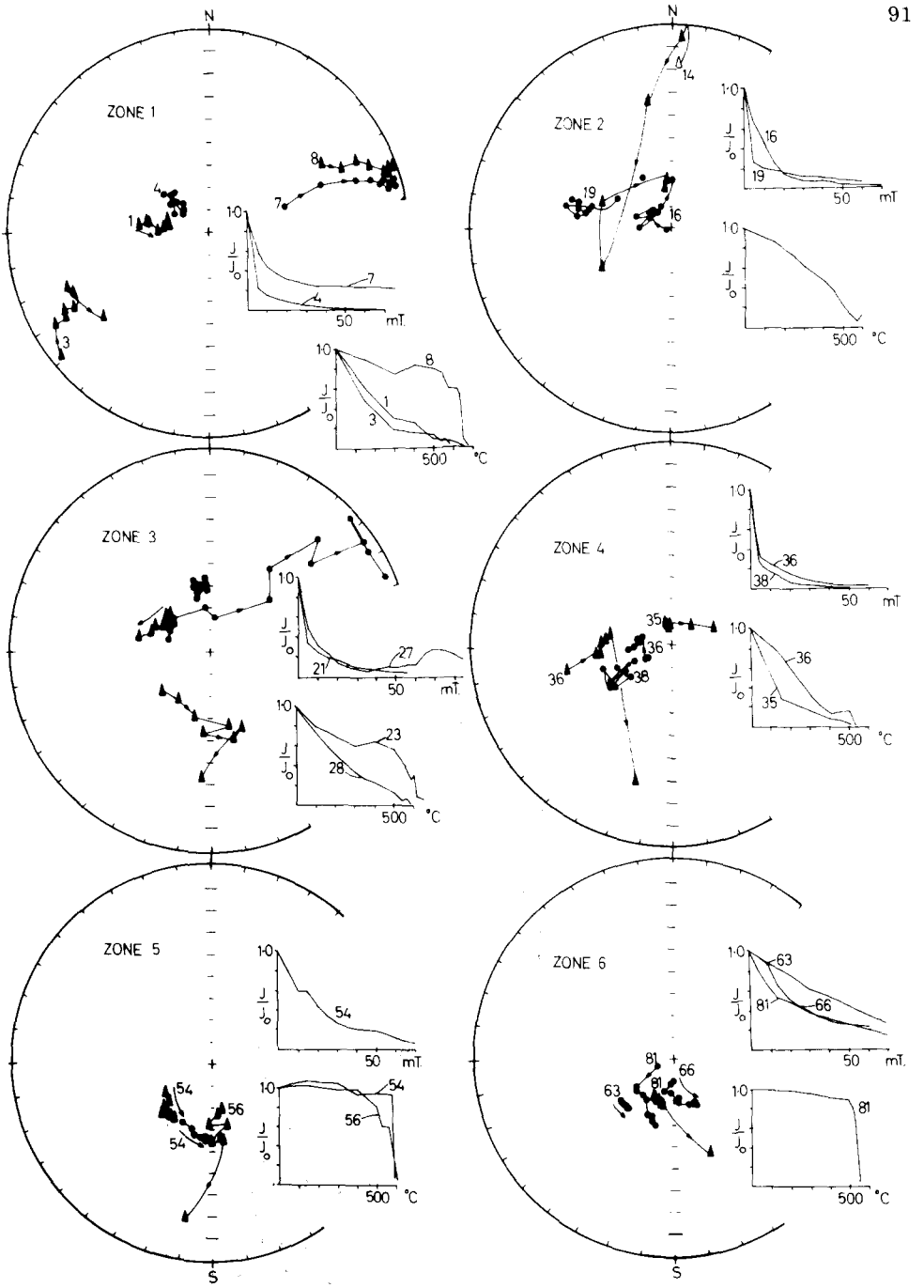


Fig. 2. Some representative results of a.f. and thermal cleaning of samples from the six zones crossing the Nagssugtoqidian belt with directional behaviour of a.f. (circles) and thermal (triangles) cleaning plotted on conventional stereographic plots. The solid symbols represent lower hemisphere plots and open symbols represent upper hemisphere plots. The decay of the natural remanence with cleaning is illustrated as a fraction of the total NRM (J_o) plotted against a.f. demagnetising field in milliTesla (mT) or cleaning temperature (°C).

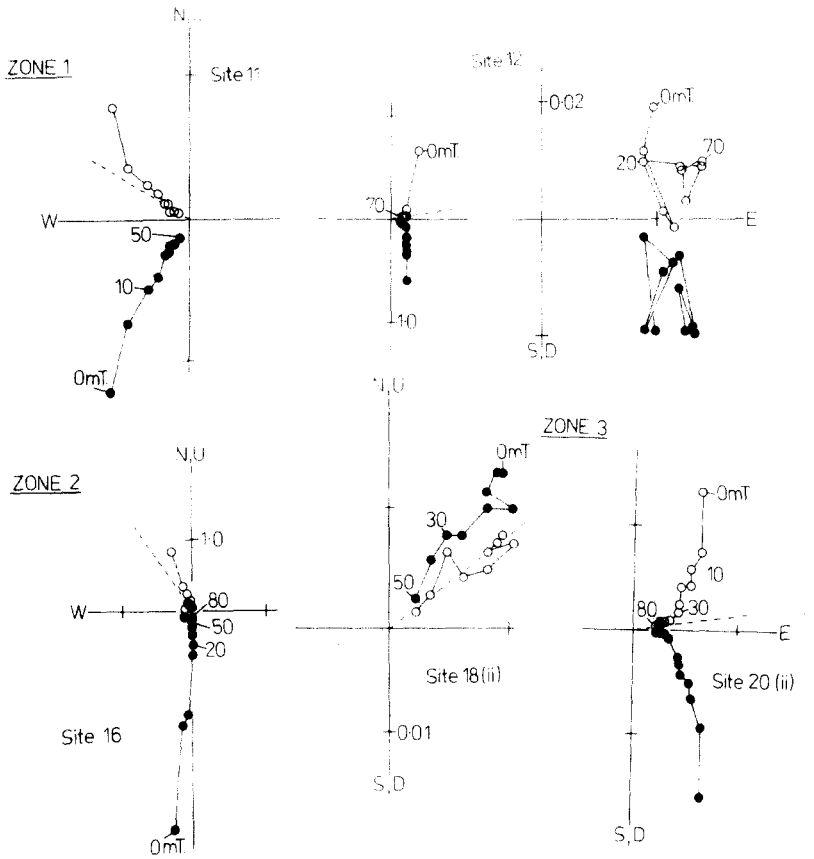


Fig. 3. Examples of routine a.f. demagnetisation of samples from zones 1 to 3 illustrated as orthogonal vector plots. Projections of the vector on to the horizontal plane are shown as open symbols and onto the vertical plane as closed symbols. Scale divisions are in 10^{-5} A.m^2 kg^{-1} and the dashed lines show the mean declination for the site. The examples from site 12 show one example with a strong low coercivity magnetite component and a weak hemo-ilmenite component, and one example with a weak hemo-ilmenite component essentially unaffected by a.f. demagnetisation.

(Fig. 2) and a minority direction, (ii) at site 16 with properties not clearly different from the remainder. At site 18 a low blocking temperature (i) is removed below 500°C but a highly-stable component (ii) in two cores remains to the Curie point of haematite.

The bulk of the coherent magnetite components in this area lie close to the younger end of the regional uplift sequence (see Fig. 5) while the remaining directions are dispersed and may have been acquired during episodic rejuvenation of the shear belt in later times; 13 (ii) and 18 (i) are comparable with the high blocking temperature component recognised in the Holsteinsborg area.

Zone 3, Ikertôq shear belt in the Sarfanguaqland region

Palaeomagnetic sites here are concentrated in amphibolitised mafic pods and lenses in quartzo-feldspathic gneisses. Two distinct components of magnetisation are recognised in these rocks. The first is the only direction present at such sites as 21, 22 and 24 where progressive fall in remanence recovers one coherent direction subtracted with thermal demagnetisation close to the Curie point of magnetite. At other sites a more complex behaviour is identified: at 20 for example, cores in group (i) show an exponential fall from high intensities to single stable end points while cores in group (ii) have much lower intensities and show little or no decrease in remanence from these weak initial intensities; one sample (Fig. 3) illustrates both components. Three other sites (23, 26, 27) also have components in low coercivity or blocking temperature ranges (Fig. 2) and isolation of the magnetite component in fields of several tens of mT is followed by migration to a shallow-directed component between the Curie points of magnetite and haematite. The magnetite-held components in this region are consistently directed N to NW with inclinations between +58 and +68 and lie just beyond the younger limit of the directional seqeunce deduced from the altitudinal studies. The hemo-ilmenite components are more variable and there is some smearing (sites 20 and 27) of sample directions in the equatorial plane.

Zone 4, Sagdlerssuaq area

The amphibolite-facies rocks are predominantly from deformed Kangamiut dykes and ultrabasic pods. Weak, moderate to low blocking temperature, components residing in magnetite predominate. Some sites (30, 34 and 38) exhibit a progressive and continuous migration of direction with treatment which is more typical of the granulite facies rocks in zone 5, but most sites show single coherent steep directions of magnetisation similar to those reported by Fahrig and Bridgwater (1976) and Beckmann and Mitchell (1976). At site 31 a low coercivity component is removed in low fields to isolate an easterly shallow direction with no further decrease in remanence. Site 33 also exhibits this behaviour, but high blocking temperature components remaining above the Curie point of magnetite are exceptional in this zone and components subtracted between 400 and 580°C are usually present (e.g., 32, 36—38). The site mean directions with three exceptions (31 (ii), 33 (ii) and 36) straddle the younger part of the directional sequence identified from the altitude sections.

Zones 5 and 6, Itivdleq Fjord

With a few exceptions, the sampling in this area has been concentrated on members of the Kangamiut dyke swarm. The hemo-ilmenite components

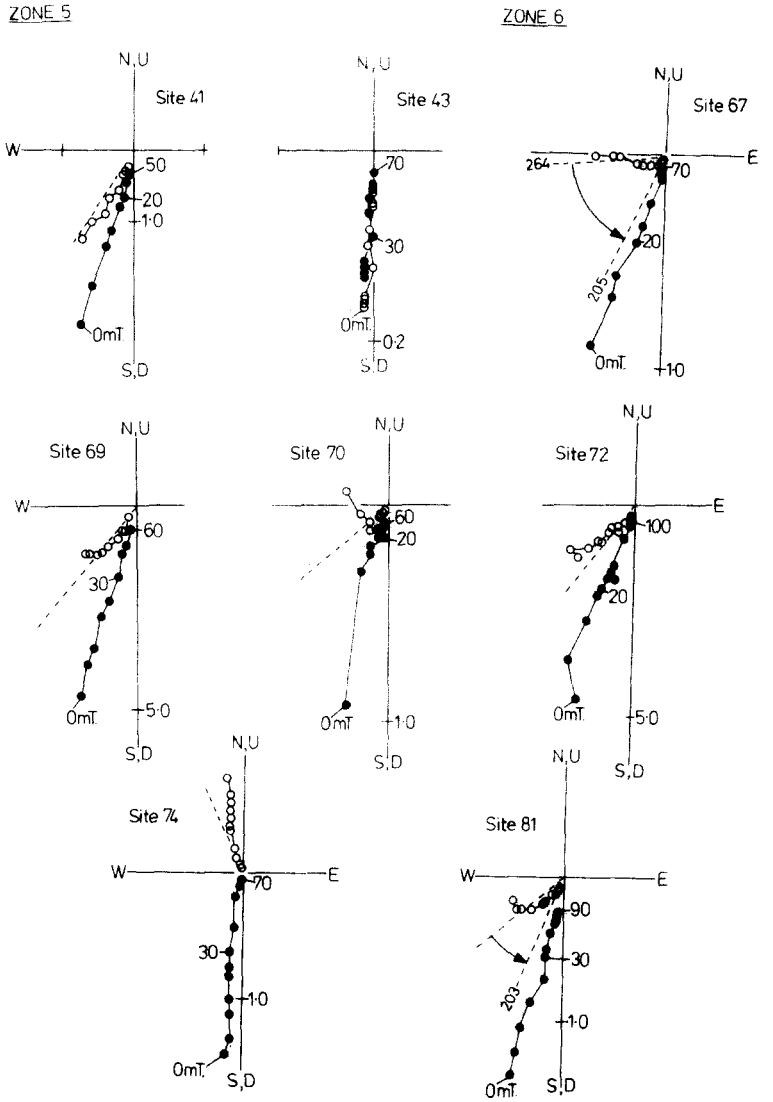


Fig. 4. Vector plots showing examples of a.f. demagnetisation from zones 5 and 6. Symbols as for Fig. 3. While some of these samples show single discrete components as in zones 1 and 2, many show a systematic anticlockwise rotation of the declination accompanied by little change in inclination. At some sites (67 and 81 illustrated here with the rotation of the mean vector before and after cleaning) this is systematically observed in several samples from the same site.

recognised in variable degrees in the northern zones are not positively recognised here. The behaviour to progressive cleaning treatment is, however, distinctive and has been described in some detail by Morgan (1976a, b). After an initial movement away from a steep positive component in very

low fields (<10 mT), sample directions typically move continuously along arcs of small circles; this movement continues to the highest stages of a.f. treatment and is reflected in the anticlockwise rotation of the declination (Fig. 4). It continues to the Curie point of magnetite (Fig. 2) and is interpreted by Morgan to represent the migration from lower to higher blocking temperature components acquired during a.p.w. motion while the terrain was slowly uplifted and cooled. This behaviour is observed to some degree in almost all of the sites in this area and the palaeofield migration direction inferred in this way is highly systematic (see Figs. 5 and 6). (There are a few cases where the migration is close to a great circle incorporating a northerly steep positive direction and in these instances ancient a.p.w. motion cannot, of course, be unambiguously separated from subtraction of a Phanerozoic component residing in a wide blocking temperature range.) Also, there are significant variations within individual sites and some cores may show systematic migration while others remain stationary. Thus although individual sites tend to be characterised by high within-site precisions due to the averaging out of secular variation over the long periods taken to cool through the blocking temperature range (Morgan, 1976a), this is not necessarily the case when a considerable variation of properties is evident within individual sites. A few sites with narrow blocking temperature ranges (e.g., 63 and 65) illustrate no systematic movement. Sites 52 and 53 are adjacent dykes, one narrow with fine grain size and the other with coarse grain size; although both sites illustrate the small circle migration each one defines a different coherent sector of the migration path (Table I). The sites in Zone 6 in the eastern part of Itivdleq Fjord have generally lower blocking temperature spectra; the lowest are observed at sites 71 and 74 with components eliminated by thermal demagnetisation below 500°C lying near the younger limit of the palaeofield migration path (Fig. 6) and site 64 which is remote from other directions. The south to north trend of oldest to youngest palaeofield directions observed by Morgan across the head of Itivdleq Fjord is also evident in the sites 40—62 of the collection between 0 and 6 km from the northern margin of Itivdleq Fjord (Fig. 6 and Table I), while in common with Morgan's three sites north of latitude 66° 37′, site 39 from the Kingâq augen reverts to an older position at the start of the palaeofield migration path. This could represent a downfaulting of the Kingâq area (Morgan, 1976b) or a pervasive mineralogical effect such that this relict granulite facies terrain passed through the critical range of blocking temperatures somewhat earlier than the amphibolitised shear zones on either side.

GEOLOGICAL AND GEOPHYSICAL IMPLICATIONS

The a.p.w. record during uplift and cooling

In conventional palaeomagnetic studies of rapidly cooled rocks it is generally possible to represent the site primary magnetisation by a single

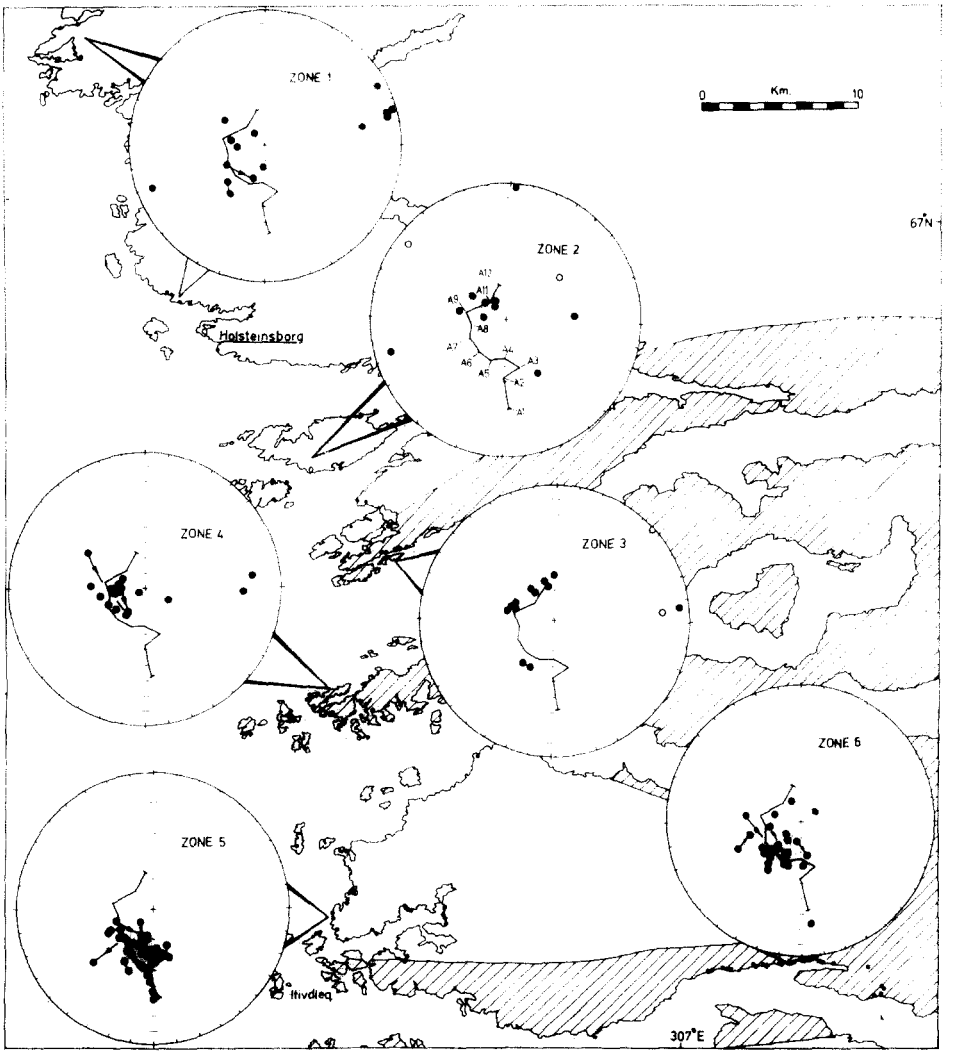
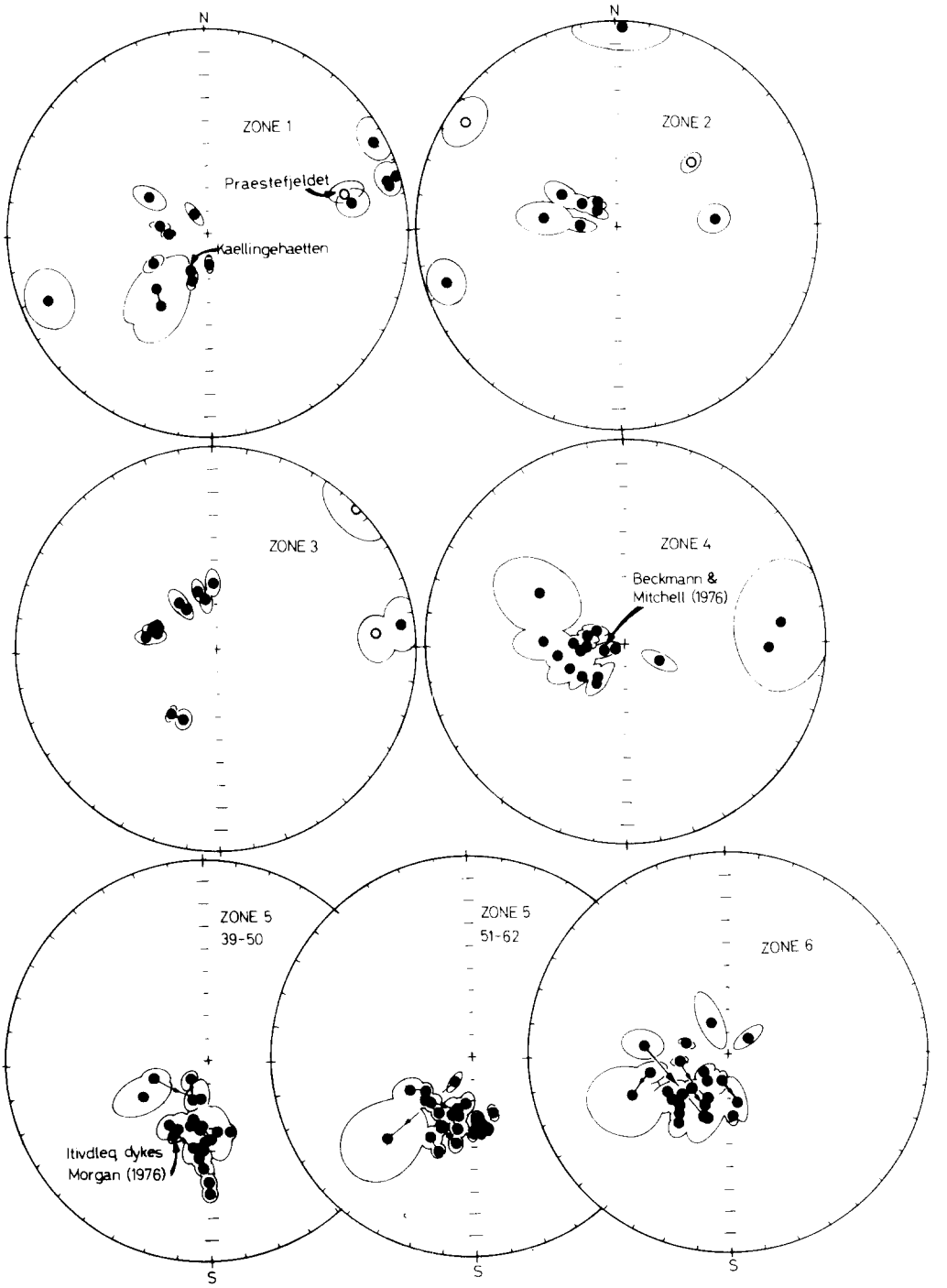


Fig. 5. The variation of the cleaned site mean directions of magnetisation area by area across the Nagssugtoqidian belt plotted on stereographic projections; other symbols are as for Fig. 1. The line on the plot illustrates the palaeofield migration path inferred from progressive demagnetisation and blocking temperature spectra of the 'A' directions (see section on a.p.w. record during uplift and cooling and Table II).

Fig. 6. All site mean directions of magnetisation after cleaning treatment with 95% confidence ovals from the six zones across the Nagssugtoqidian belt plotted on stereographic projections. The results of earlier palaeomagnetic studies within the Nagssugtoqidian belt are also shown. The arrows indicate the direction of motion from lower to higher blocking temperature or coercivity components.


N
ZONE 1
Praestefjeldet
Kaellingehaetten
N
ZONE 2
ZONE 3
ZONE 4
Beckmann &
Mitchell (1976)
ZONE 5
39-50
Itivdleq dykes
Morgan (1976)
ZONE 5
51-62
ZONE 6
S
S
S

direction, or in favourable circumstances by more than one co-existing direction residing in different blocking temperature ranges. In slowly-cooled rocks this situation is displaced by a line segment where individual samples and sites record a discontinuous record of the palaeofield migration during cooling. In the Nagssugtoqidian terrain this migration is defined in part by three independent and complementary methods. Firstly there is the trend in palaeomagnetic directions in altitude sections which, on the basis of the assumption that higher topography cooled prior to lower topography, define a trend from earlier to later magnetisations (Piper, 1981), reflected in a clockwise movement of the palaeofield vector and a movement from shallower to steeper inclinations (Fig. 5). Secondly, this trend is identified by the migration of directions with progressive cleaning in many samples with distributed blocking temperature ranges (notably in zones 5 and 6). Thirdly, there is a general tendency for sites with steep westerly directions of magnetisation lying near the younger part of the palaeofield migration path to have low blocking temperature spectra (notably in zones 3 and 4) and sites with shallower southerly magnetisations lying near the older part of the palaeofield migration path have high blocking temperature spectra (notably in zone 5). Thus the palaeofield migration path defined by the altitude sections (Fig. 6) is uniformly supported by data from all six zones across the Nagssugtoqidian belt (Fig. 5) and can confidently be interpreted as recording the signature of the a.p.w. path during cooling of this exposed part of the terrain (Morgan, 1976b).

In reality the part or parts of the a.p.w. path recorded by any one sample or site will depend on the nature of the magnetic phases, and hence indirectly on the chemistry and cooling history of the rock unit, and several examples have been recognised in the field study of diverse behaviour in similar or closely spaced units. Thus the complete history of a.p.w. during cooling must be obtained from the overall site mean data: there are two clear components, namely the clockwise declination and shallow to steep inclination movement resolved from the altitude sections grouped as the 'A' magnetisations, and the ENE—WSW axis of dual polarity and shallow inclination residing in the highest blocking temperature components from the northern part of the belt and grouped as the 'B' magnetisations. To compute representative palaeomagnetic poles for the overall migration, the total data are subdivided into approximately 20° × 20° sectors of the stereonet (Tables II and III). The main migration from a palaeofield direction $D = 174°$, $I = 32°$ to $D = 348°$, $I = 66°$ is then represented by twelve palaeomagnetic poles which incorporate all but five of the lower blocking temperature site mean directions (probably all magnetite-held, see Table I). Although these subdivisions are somewhat arbitrary, they comprise remarkably uniform distributions and only two sites in the A7 group and one site in the A10 group are responsible for distribution of inclinations exceeding the 18—20° range of all the remaining groups. The resolution of only one polarity in 161 magnetisations along this path implies that one polarity of

TABLE II

Group mean calculation for palaeopoles falling in the 'A' Group defining the SE—NW a.p.w. Swathe

Pole Reference	Sites included in calculation	N	R	D	I	k	Palaeomagnetic Pole °E	°N	dp/dm	Pole relative to Laurentian Shield	
A1 (D=166-181, I=24-39)	43(i), 43(ii), 77(ii), 32(i)*, 32(ii)*	5	4.96	174.3	32.3	96	312	5	5/8.9	295	1
A2 (D=172-188, I=42-58)	39, 40(i), 44(ii), 45(i), 45(ii), 46(i), 56(i), 56(ii), 61, 30*	10	9.94	180.5	52.9	151	306	10	4/5	290	8
A3 (D=148-168, I=49-64)	48, 60(i), 60(ii), 16(ii), 11(ii)*, 18*	6	5.96	161.6	59.0	130	322	18	7/9	306	14
A4 (D=170-190, I=60-76)	6, 40(ii), 44(i), 53(i), 65, 66(ii), 61(i), 1(i)*, 1(ii)*, 2(ii)*, 11(i)*, 13(ii)*	12	11.95	181.2	65.3	227	306	24	4/5	291	21
A5 (D=190-209, I=51-74)	2(ii), 23(ii), 41(ii), 46(ii), 42(ii), 47, 49(ii), 50(ii), 51(i), 51(ii), 52(ii), 53(ii), 58(ii), 59(i), 59(ii), 62(i), 62(ii), 66(i), 67(ii), 76(ii), 79, 81(ii), 2(i)*, 3(ii)*, 4*, 12(ii)*, 13(i)*, 20*, 22*, 28*, 31*, 33*, 39*.	33	32.7	198.7	62.9	107	292	22	3/4	277	20
A6 (D=212-230, I=51-72)	3(i), 3(ii), 30(ii), 32(ii), 41(i), 42(i), 52(i), 54(i), 54(ii), 55(i), 55(ii), 63, 69(i), 69(ii), 72(ii), 75(ii), 87(i), 78, 81(i), 3(i)*, 5(ii)*, 6(ii)*, 9(i)*, 9(ii)*, 10(i)*, 10(ii)*, 12(i)*, 15(i)*, 15(ii)*, 19(i)*, 19(ii)*, 21*, 26*, 27*	34	33.6	220.6	63.4	83	276	26	3/4	262	26

Pole Reference	Sites included in calculation	N	R	D	I	k	Palaeomagnetic Pole °E	°N	dp/dm	Pole relative to Laurentian Shield	
A7 (D=230-249, I=44-86)	2(i), 31(I), 37, 38(ii), 50, 57(i), 57(ii), 68(ii), 70, 72(i), 75(i), 5(i)*, 6(i)*, 14(ii)*, 14(ii)*, 23*, 25(ii)*, 29*	18	17.77	238.9	64.1	73	263	31	5/6	250	32
A8 (D=254-268, I=52-73)	1, 33(i), 35, 38(i), 49(i), 67(i), 68(i), 7(i)*, 7(ii)*, 8(i)*, 8(ii)*, 17(i)*, 24(ii)*	13	12.91	260.4	66.0	128	248	40	5/6	236	42
A9 (D=271-288, I=52-74)	5, 13(i), 15(ii), 23(i), 25, 28, 29, 30(i), 34(i), 34(ii), 71, 76(i), 24*	13	12.86	279.3	63.6	88	231	44	6/7	219	48
A10 (D=292-303, I=46-77)	11(ii), 19(i), 24, 32(i), 58(i), 17(ii)*	6	5.92	299.5	61.8	62	211	50	10/13	198	55
A11 (D=304-326, I=63-80)	4, 14, 16(i), 17, 22, 27, 30(ii)*	7	6.96	318.2	73.4	155	214	70	8/9	208	75
A12 (D=334-358, I=60-74)	20(i), 21, 26(i), 74	4	3.98	348.1	66.2	127	151	71	11/13	121	75

Symbols as for Table I: Site numbers indicated by asterix are from altitude sections in the same terrain and reported in Piper (1981).

TABLE III

Group mean calculations for palaeopoles falling in the 'B' Group

Pole Reference	Sites included in calculation	N	R	D	I	k	Palaeomagnetic Pole °E	°N	dp/dm	Pole relative to Laurentian Shield	
B1 (D=71-95, I=-10 to -28)	13(ii), 26(ii), 34(ii), 64, 36*, 38*, 34(ii)*	7	6.85	82.1	-21.4	40	48	-7	5/10	32	-11
B2 (D=75-83, I=2 to -19)	7, 8, 10, 20, 31(ii)	5	4.97	78.8	8.0	117	46	8	4/7	29	4
B3 (D=86-92, I=25 to 46)	12, 19(ii), 33	3	2.99	86.0	25.1	222	35	14	5/9	17	9
B4 (D=62-70, I=5 to -12)	3(iii), 9, 34(i)*, 35(i)*	4	3.95	64.8	-9.0	64	62	5	6/12	45	2
B5 (D=46-57, I=-1 to -47)	11(i), 18, 27	3	2.84	48.4	-24.7	12	80	3	21/39	63	2

Symbols as for Tables I and II; underlined sites give antiparallel directions and have been reversed for the group mean calculation.

the geomagnetic field was strongly and persistently dominant during slow cooling. The 'B' directions (Table III) probably indicate the migration of a dual polarity axis from $D = 82°$, $I = -24°$ to $D = 48°$, $I = -25°$ because the former direction is dominant in the higher altitude sites at Praestfjeldet, while the latter direction is present in low altitude sites at sea level around Holsteinsborg.

Although slowly-cooled plutonic terrains are providing the most complete record of a.p.w. movements (and complementary continental movements) in Precambrian times, it is generally difficult to assign a precise age to the magnetisations. In the Nagssugtoqidian belt the magnetite-held components are likely to be somewhat older than argon closure in biotites indicated by ages of 1790—1650 Ma (Larson and Møller, 1968), but comparable to the Rb—Sr whole rock/mineral ages of 1740—1600 Ma (M.H. Hickman personal communication, 1983). K—Ar isochrons of 1830 and 1700 Ma at Itivdleq and 1620 Ma on Sagdlerssuaq may reflect the difference in magnetisation age of the two areas which lie on older and younger parts of the palaeofield migration path, respectively. The magnetic blocking temperatures are certainly higher than the argon blocking temperatures, but they may not be much older in this terrain because the late tectonic events and granulite-facies metamorphism are dated 1750 ± 100 Ma. This defines a stage when the terrain in the north, at least, was still deeply buried and above the Curie points of both haematite and magnetite (Davidson, 1979). Thus the overall path may represent no more than a few tens of millions of years of a.p.w. motion.

The palaeopoles from the Nagssugtoqidian belt are summarised in Fig. 7

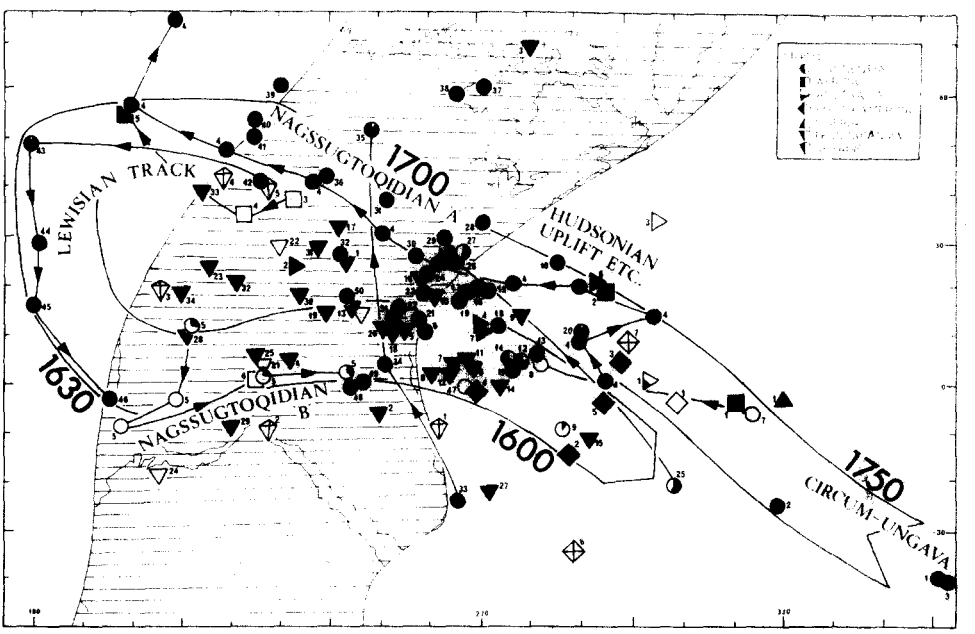


Fig. 7. Palaeomagnetic south poles assigned to the interval c. 1750—1600 Ma on the Proterozoic Supercontinent reconstruction (Piper, 1982). The Laurentian Shield is retained in present coordinates and other shields and poles are rotated towards it. The palaeopoles are: 1, Richmond Gulf Volcanics; 2, Richmond Gulf Manitounuk Island Volcanics; 3, Eskimo Volcanics (1750*—1625^{+}); 4, Sequence of 'A' uplift magnetisations, Nagssugtoqidian terrain (1790—1650^{+}); 5, Sequence of 'B' uplift magnetissations, same terrain (1690*); 6, Daly Bay metamorphics 'A' (1690—1590^{+}); 7, Pearson Formation 'B'; 8, Pearson Formation 'C'; 9, Martin Formation (1830—1650^{+}); 10, Kaellinghaetten gneiss, W. Greenland; 11, Retty peridotite, Labrador 'B'; 12, Menihek Formation, Labrador 'B'; 13, Big Spruce Complex 'Y'; 14, Dubawnt Group (1760*0; 15, Itivdleq dykes (1750^{+}); 16, Coronation geosyncline overprint; 17, Metamorphosed Kaminak dykes (1900—1620^{+}); 18, Nanacho Group (1850—1750^{+}); 19, Kahocella secondary; 20, Sparrow dykes (1550—1390^{+}, 1700^{++}); 21, Sokoman iron formation, post-folding remanence; 22, Itivdleq dykes (1790—1650^{+}); 23, Kangamiut dykes south of Nagssugtoqidian Front; 24, Amitsoq gneisses; 25, Flin-Flon secondary (1760*); 26, Flin-Flon Tertiary (1700—1600^{+}); 27, Gunflint Formation (1600*); 28, Spanish River Carbonatite (1750*); 29, Nordre Strømfjord gneisses; 30, Angmagssalik gneisses (1800—1600^{+}); 31—32, Molson dykes (1650—1280^{+}); 33, Seward Formation 'A'; 34, Seward Formation 'B'; 35, Seward Formation 'C'; 36, Wind River dykes (1880—1680^{+}); 37, Sagdlerssuaq dykes 1620^{++}); 38—39, Ketilidian metavolcanics (1700—1600^{+}); 40, Amphibolitised dykes, north of Nagssugtoqidian Front; 41, Sarfanguaqland amphibolites; 42, Lewisian 'A2' (1650—1625^{+}); 43, Lewisian 'A3' (1690—1680^{+}); 44, Lewisian 'A4' (1690—1670^{+}); 45, Lewisian 'A5' (1500—1470^{+}); 46, Lewisian 'B' (1630*); 47, Beartooth dykes, 7 and 8 (1500^{+}); 48, Flaherty Volcanics (47, 1690^{+}—1640*); 49, Haig intrusions (1790*—1620^{+}); 50, Eskimo Volcanics, secondary; 51, Cape Smith Metavolcanics (1650—1450^{+}). AFRICAN SHIELD: 1, Dykes intruding Gueld el Hadid Series (1760—1350^{+}); 2, Dykes intruding d'Akileb Deilel Series (1760—1350^{+}); 3—5, Waterberg Formation, higher sites (1760** Ma); 6, Van dyke Mine, dolerite dyke, South Africa (1615*). SOUTH AMERICAN SHIELD: 1, Cudivero-Caicara, acid metavolcanics (1960—1640*); 2, Roraima Sandstone (1660*); 3—4, Roraima intrusive suite (1660*); 5, Paraguaza pluton (1590*—1524**,$^{+}$); 6, Telles Pires (1610*); 7, Beneficente Group (1480*). FENNOSCANDIAN SHIELD: 1—2, Vuollerim gabbro; 3, Hame dolerites (1800—1640*); 4, Radmanso gabbro—diorite; 5, Tarendo Gabbro; 6, Tarendo acid rocks; 7, Ylivieska Gabbro; 8, Pohjanmaa gabbro—diorite; 9, Tammela gabbro—diorite; 10, Mikkeli gabbro—diorite; 11, Hyvinkaa gabbro-diorite; 12, SW Finland gabbro—diorites; 13—16, Kallax Gabbro; 17, Korstrask Gabbro; 18, Niemisel Gabbro; 19, Sangis Gabbro; 21—24, Gallivare basic rocks; 25—26, Stora Lulevatten gabbro; 27—28, Jokkmokk basic rocks; 29, Kumlinge dolerites (1640**); 30, Loftahammar gabbro (1660—1470*$^{+}$); 31, Uppsala metabasites 'A' (1570*); 32, Foglo dolerite; 33, Ava intrusive (1700—1670*); 34, Upper Dala Volcanics (1634*). AUSTRALIAN SHIELD: 1, YD dykes (1700); 2, YC dykes (1500, <YD); 3, GB dykes (1670*); 4, GA dykes (1470*); 5, Iron Monarch, negative group ore (1750—1510); 6, Iron Prince ore (1750—1510); 7, Gawler Range Volcanics (1525*). SIBERIAN SHIELD: 1, Diabase and diorite, Aldan region (1590^{+}); UKRAINIAN SHIELD: 1, Uman granite; 2, Turchinka gabbro; 3, Korosten gabbros (1550^{+}—1650^{+}); 4, Ukrainian diabases (1550^{+}—1650^{+}); 5, Olivine diabases and lamprophyres (1770^{+}). ($^{+}$K—Ar, $^{++}$ ^{39}Ar: ^{40}Ar or K—Ar isochron, *Rb—Sr, **U—Pb age). The zone of horizontal shading is the area of the continental crustal in lower and middle Proterozoic times covered by the coordinates of this diagram and shows the continental outlines of Greenland, North America, Africa and South America. For further details of the reconstruction and the palaeomagnetic evidence see Piper (1982).

with contemporaneous data from the remainder of the Laurentian Shield, after closure of the Davis Strait according to the conventional reconstruction of Bullard et al. (1965). These c. 1750—1600 Ma data are immediately preceded by the palaeomagnetic record from the Coronation geosyncline, Authapuscow aulacogen and Kilohigol basin which developed around the margins of the Slave craton in late Aphebian (c. 2100—1700 Ma) times. These latter data define in stratigraphic sequence an elongated a.p.w. loop, the "Coronation Loop" of McGlynn and Irving (1978), and palaeopoles from the highest formations in the succession (Douglas Peninsula and Pearson Formation) define part of a hairpin at c. 1750 Ma (Fig. 7). These youngest poles from the circum-Slave basins are similar to poles from a range of Aphebian rocks in the Labrador trough which, however, are probably all metamorphic magnetisations. Other poles from late Aphebian rocks of the circum-Ungava geosyncline (Richmond Gulf and Belcher Island) in Labrador fall in the central-southern Atlantic some 30° SE of the youngest part of the Coronation path and the oldest part of the Nagssugtoqidian path; they include magnetisations dated 1750 Ma (Schmidt, 1980) and appear to define an extension to the a.p.w. paths as previously defined by Irving and McGlynn (1979) at this point. Magnetisations from the circum-Slave successions, considered to be secondary, are identified in the Pearson Formation (McGlynn and Irving, 1978) and an extensive secondary overprint is present in these successions (Evans and Haye, 1981) coincident with the oldest part of the Nagssugtoqidian track. This uplift and overprint record in the Canadian Shield is linked to the widespread c. 1700 Ma K—Ar dates and attributed to the "Hudsonian" mobile activity in the Churchill, Bear and other adjoining terrains. In addition to a thermal overprint, magnetisations comparable to the Nagssugtoqidian record are present in those terrains elsewhere in the Laurentian Shield which were directly affected by the c. 1800 Ma mobile activity: Park (1975) determined two magnetisations in amphibolite terrain of the Flin-Flon belt in Manitoba and Saskatchewan and suggested in outline the SE—NW trend in the a.p.w. track defined in detail from Greenland (Morgan, 1976 and this study); the same track was also recognised by Park (1977) in metamorphic magnetisations of the Labrador trough. In addition to the geochronologic evidence from Greenland, an assignment of c. 1700 Ma for the oldest part of the Nagssugtoqidian track comes from Ar^{39}/Ar^{40} studies of the Sparrow dykes (McGlynn et al., 1974).

The continuation of the a.p.w. path as defined from the Nagssugtoqidian terrain is defined by only very limited data from North America but is represented in part (Fig. 7) by the magnetisation record from the Lewisian terrain of NW Scotland (Smith and Piper, 1982). Clearly the segments of an a.p.w. path recovered from any one area depend critically on the level of exposure in the terrain: in the Nagssugtoqidian belt the maximum thickness of crust studied is ~ 1500 m in a rock section which has been uplifted from a depth of the order of 20 km. Only by wider areal study could the full a.p.w. record be recovered.

The highest blocking temperature components yielding the shallow ENE—WSW axis are somewhat more difficult to assign to a specific part on the a.p.w. path. They had been allocated to an extremity of the Coronation loop defined by the Stark and Totachwi (circum-Slave) poles and the high blocking temperature components recognised in relict terrain in the Lewisian Complex in NW Scotland (Piper, 1982). However, since the late tectonic diorites are evidently no older than ~ 1700 Ma and this axis of magnetisation is present only in the northern part of the Nagssugtoqidian belt where the metamorphic facies are younger and the Nagssugtoqidian reworking is profound, this assignment must be incorrect. It now seems likely that these haematite—ilmenite components actually record a *later* magnetisation than the predominant magnetite-held 'A' components because in positional terms the path defined by the 'B' poles falls along the younger part of the Lewisian track incorporating magnetisations dated 1630 Ma and a range of Fennoscandian poles dated 1660—1600 Ma (Fig. 7). This would be consistent with magnetisation of the diorites after intrusion at 1750—1690 Ma and uplift from great depths after this. In general the assignment of high blocking temperature haematite-held remanences to older or younger ages than co-existing magnetite-held remanences is ambiguous because the haematite may have a primary magmatic, or secondary chemical, origin (Buchan and Dunlop, 1976).

Collectively the accordance of the a.p.w. path from the Nagssugtoqidian terrain with 1750—1600 Ma data from other major shields and in particular with uplift metamorphic magnetisations from terrains mobilised during the c. 1800 Ma episodes, supports age assignments of ~ 1760—1650 and 1630—1600 Ma, respectively, for the 'A' and 'B' groups of palaeopoles in Tables II and III and Fig. 7. They define uplift and cooling of this terrain to c. 600—500°C after late stages of the Nag 2 episode at c. 1700 Ma; in common with the Rb—Sr and K—Ar data they are consistent with a relatively rapid uplift of the terrain after the c. 1800 Ma "Hudsonian" mobile episode.

It remains to discuss the nature of the progressive migration of the magnetic vectors observed in the relict Archaean terrain of the Itivdleq area (Morgan, 1976a, b). In principle, this could represent either the progressive recovery of a single continuous high blocking temperature spectrum (as interpreted by Morgan), or it could represent the subtraction of a secondary component and the incomplete recovery of a primary component of late Archaean/early Proterozoic age and with similar spectrum. In the light of this wider areal study, Morgan's interpretation remains the most probable one for several reasons. Firstly, the migrations lie along an identical path to the one recovered from the terrains profoundly reworked at 1800 Ma and controlled by altitude sections and blocking temperature relationships. Secondly, the remanence has been shown to reside in a discrete exsolution phase or submicron size magnetite grains within plagioclase feldspars of single or pseudo-single domain size (Morgan and Smith, 1981). Thirdly,

the migrations do not begin and end close to definitive end points but span variable sectors of a large part of the overall path. Fourthly, the K—Ar uplift ages are derived in part from this area, and the terrain immediately to the north was at profound depths and at temperatures well in excess of the Curie temperatures prior to 1800 Ma. Fifthly, geological evidence shows that the regional gneisses were at metamorphic temperatures when the 1950 Ma Kangamiut dykes were intruded (Watterson, 1974). For these reasons it seems legitimate to interpret (as is done here) the entire remanence record in the context of post-Nagssugtoqidian uplift and cooling.

Regional variations in magnetisation history

Variations in palaeomagnetic direction from area to area across the Nagssugtoqidian belt (Figs. 5 and 6) illustrate the complexity of the palaeomagnetic record within a slowly-cooled terrain; they emphasise the importance of widespread regional samples for evaluating the migration of the magnetic field vector during the interval of uplift and cooling. On a local scale it is not always clear, however, whether these differences are explicable in terms of blocking temperature variations, thermal "highs" and "lows", or include a component of differential vertical movement. On a regional scale the whole exposed level of the Nagssugtoqidian belt between 66.5 and 67.3°N cooled through the critical blocking temperature range at 1750—1600 Ma but some important regional variations are also present. These include the high blocking temperature component, apparently postdating the lower blocking temperature components to the north and south, recorded by hemo-ilmenite phases in the diorites at sea level around the town of Holsteinsborg, and partially up the topography on the northern margin of the belt. This might indicate a thermal dome representing a later reheating event, or be due to uplifting of a segment of terrain between Kaellinghaetten and Kangardluarssuaq to the north (Fig. 8). The 'B' directions are not unique to the late tectonic diorites because these rocks further to the north (sites 1 to 6 and Piper (1981)) yield directions falling on the 'A' magnetisation path (i.e., the magnetisations are c. 1700 Ma in age and close to the intrusion age of the diorites).

In zone 2 the data fall near the younger part of the palaeofield migration path but include a diverse range of magnetisations suggesting a more complex history than elsewhere and a possible link to the structural complexities and presence of pseudotachylites in this area (Bak et al., 1975). Zone 3 data fall at the youngest section of the path and indicate a regional "tilt" to the north across this margin of the shear belt. The reverse tilt is indicated by the palaeomagnetic data across the southern margin in zones 4 and 5. The remanence directions from Nordre Strømfjord fall along the early part of the 'A' track (Beckmann et al., 1977). This is the regional effect of a thermal dome with its axis near the northern margin of the Ikertôq shear belt (Fig. 8) but the contribution of vertical fault motions to this regional picture is unclear.

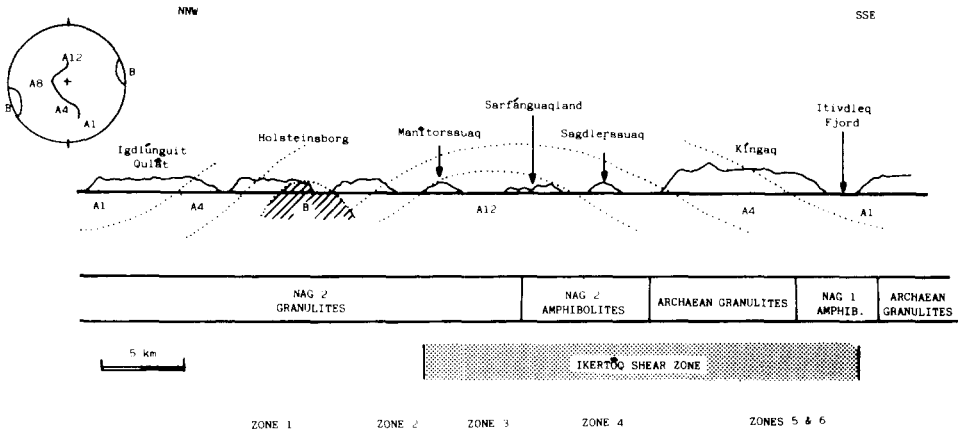


Fig. 8. Diagrammatic NNW—SSE cross section through the Nagssugtoqidian mobile belt showing the change in the predominant magnetite-held remanence as judged from position along the a.p.w. swathe. These would represent the state of uplift-related cooling across the belt through c. 600—500°C c. 1750—1650 Ma. The 'B' remanences in the Holsteinsborg area are believed to represent a slightly later reheating event of local significance. Note that the model of a thermal dome as suggested here cannot be distinguished from the effects of post-1600 Ma relative vertical movements along faults running broadly parallel to the belt and such a model could provide a second possible scenario for interpretation of the palaeomagnetic data.

Magnetic properties

In Fig. 9 the total NRM intensities and the low field magnetic susceptibilities of the palaeomagnetic cores are shown as normalised distributions after Irving et al. (1966). Granulites and amphibolites show peaked distributions of both parameters but there is no systematic tendency for granulites to be more strongly magnetised than amphibolites from Nagssugtoqidian and/or relict Archaean terrain. This probably reflects two observations: firstly the sampled amphibolites are of high grade and in part retrogressed granulite facies (Korstgaard, 1979); in lower grade amphibolite and greenschist facies the iron tends to be variably incorporated into the metamorphic silicates, especially amphibole and biotite, at the expense of magnetite. Secondly the high stability remanence resides in part in haematite phases with lower remanence and susceptibility than magnetite which predominates in the amphibolites. The only clear tendency is for the dykes to be more strongly magnetised than the country rocks (Fig. 9). It has been suggested (e.g., Watson, 1973) that strongly magnetised and dense areas of Archaean gneiss terrains represent relict granulites which have escaped later reworking. This may be the case in some terrains but is not apparently evident in the sample properties from this area; a regional magnetic survey would, however, be necessary to clarify this point.

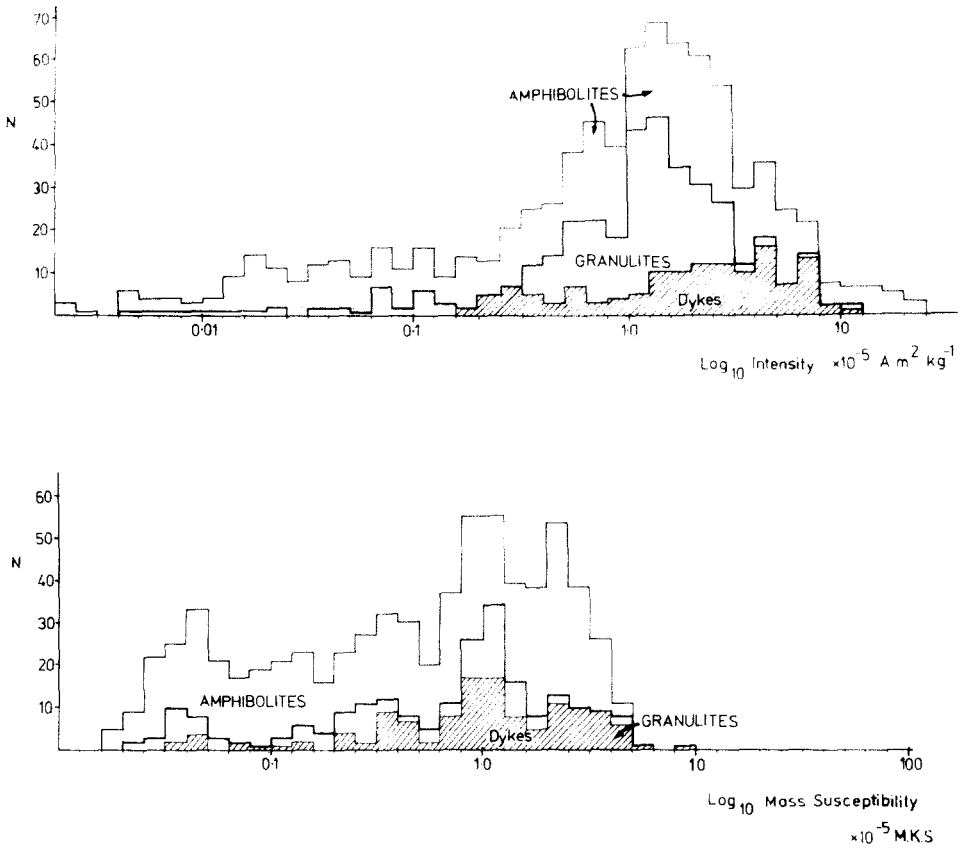


Fig. 9. Normalised Remanent Magnetisations and mass susceptibilities of the palaeomagnetic cores (not all cores were measured for susceptibility). The samples have been divided into the two broad categories of those falling in amphibolite terrain and those falling in granulite terrain. The dyke samples in granulite terrain are distinguished from the remaining country rock samples but no further subdivisions are attempted because there are no other significant distributions in these collections.

ACKNOWLEDGEMENTS

This work was financed by Natural Environment Research Council Grant GR3/2398, and logistical support was provided by the Liverpool Precambrian Boundary Program directed by Dr. J.S. Watterson. I am very grateful to Dr. Watterson for valuable discussions, Dr. R.L. Smith for assistance with the fieldwork, Mrs. Joan Dean for making the a.f. demagnetisation measurements, and Professor Martha Hickman for valuable information on the radiometric studies and for suggesting many improvements to the manuscript. We are grateful to the Greenland Ministry for permission to work in Greenland.

REFERENCES

Bak, J., Sorensen, K., Grocott, J., Korstgaard, J.A., Nash, D. and Watterson, J., 1975. Tectonic implications of Precambrian shear belts in western Greenland. Nature, 254: 566—569.

Beckmann, G.E.J., 1979. A review of Nagssugtoqidian palaeomagnetism. Grøn. Geol. Unders., Rapp., 89: 115—124.

Beckmann, G.E.J., 1983. Palaeomagnetism of some Precambrian rocks in south-east Greenland. Phys. Earth Planet. Inter., 32: 85—99.

Beckmann, G.E.J. and Mitchell, J.E., 1976. Palaeomagnetic and geochronological work in central West Greenland. Earth Planet. Sci. Lett., 30: 269—280.

Beckmann, G.E.J., Olesen, N.Ø. and Sorensen, K., 1977. A palaeomagnetic experiment on crustal uplift in West Greenland. Earth Planet. Sci. Lett., 36: 269—279.

Bridgwater, D., Watson, J. and Windley, B.F., 1973. The Archaean craton of the North Atlantic region. Philos. Trans. R. Soc. London, Soc. A: 273: 493—512.

Buchan, K.L. and Dunlop, D.J., 1976. Palaeomagnetism of the Haliburton intrusions: superimposed magnetisations, metamorphism and tectonics in the late Precambrian. J. Geophys. Res., 81: 2951—2967.

Bullard, E.C., Everett, J.C. and Smith, A.G., 1965. The fit of the continents around the Atlantic. Philos. Trans. R. Soc. London Ser. A: 258: 41—51.

Dagley, P. and Ade-Hall, J.M., 1970. Cretaceous, Tertiary and Quaternary palaeomagnetic results from Hungary. Geophys. J.R. Astron. Soc., 20: 65—87.

Davidson, L.M., 1979. Nagssugtoqidian granulite facies metamorphism in the Holsteinsborg region, West Greenland. Grøn. Geol. Unders., Rapp., 89: 97—108.

Davidson, L.M. and Park, R.G., 1978. Late Nagssugtoqidian stress orientation derived from deformed granodiorite dykes north of Holsteinsborg, West Greenland. Geol. Soc. London, 135: 283—289.

Dodson, M.H. and McClelland-Brown, E., 1980. Magnetic blocking temperatures of single-domain grains during slow cooling. J. Geophys. Res., 85: 2625—2637.

Doell, R.R. and Cox, A., 1967. Palaeomagnetic sampling with a portable drill. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in Palaeomagnetism. Elsevier Publ. Co., Amsterdam, pp. 21—25.

Dickinson, B.B. and Watson, J.V., 1976. Variations in crustal level and geothermal gradient during the evolution of the Lewisian Complex of NorthWest Scotland. Precambrian Res., 3: 363—374.

Escher, A., Jack, S. and Watterson, J., 1976. Tectonics of the North Atlantic Proterozoic dyke swarm. Phil. Trans. R. Soc. Lond. A, 280: 529—539.

Evans, M.E. and Hoye, G.S., 1981. Palaeomagnetic results from the Lower Proterozoic rocks of Great Slave Lake and Bathurst Inlet areas, N.W.T. In: F.H.A. Campbell (Editor), The Proterozoic Basins of Canada. Geol. Surv. Can. Pap. 81-10., pp. 16—32.

Fahrig, W.F. and Bridgwater, D., 1976. Late Archaean—Early Proterozoic palaeomagnetic pole positions from West Greenland. In: B.F. Windley (Editor), The Early History of the Earth. Wiley and Sons, London, pp. 427—439.

Grocott, J., 1979. Controls of metamorphic grade in shear belts. Grøn. Geol. Unders., Rapp., 89: 47—62.

Hart, S.R., 1981. Diffusion compensation in natural silicates. Geochim. Cosmochim. Acta, 45: 279—291.

Hickman, M.H., 1979. A Rb—Sr age and isotope study of the Ikertôq, Nordre Strømfjord, and Evighedsfjord shear belts, West Greenland — outline and preliminary results. Grøn. Geol. Unders., Rapp., 89: 125—128.

Irving, E. and McGlynn, J.C., 1979. Palaeomagnetism in the Coronation Geosyncline and arrangement of continents in the middle Proterozoic. Geophys. J. R. Astron. Soc., 58: 309—336.

Irving, E., Molyneux, L. and Runcorn, S.K., 1966. The analysis of remanent magnetisation intensities and susceptibilities of rocks. Geophys. J. R. Astron. Soc., 10: 451—464.

Kaalsbeck, F., Bridgwater, D. and Zeck, H.P., 1978. A 1950 (± 60) Ma Rb—Sr whole rock isochron age from two Kangamiut dykes and the timing of the Nagssugtoqidian (Hudsonian) orogeny in West Greenland. Can. J. Earth Sci., 15: 1122—1128.

Korstgaard, J.A., 1979. Metamorphism of the Kangamiut dykes and the metamorphic and structural evolution of the southern Nagssugtoqidian boundary in the Itivdleq-Ikertôq region, West Greenland. Grøn. Geol. Unders., Rapp., 89: 63—76.

Larson, O. and Møller, J., 1968. K/Ar age determinations from Western Greenland, I. Reconnaissance programme. Grøn. Geol. Unders., Rapp., 15: 82—86.

McGlynn, J.C. and Irving, E., 1978. Palaeomagnetism of the Pearson Formation, Great Slave Supergroup, N.W.T. and the Coronation Loop. Can. J. Earth Sci., 15: 642—654.

McGlynn, J.C., Hanson, G.N., Irving, E. and Park, J.K., 1974. Palaeomagnetism and age of Nonacho Group sandstones and associated Sparrow dykes, District of Mackenzie. Can. J. Earth Sci., 11: 30—42.

McWilliams, M.O. and Dunlop, D.J., 1978. Grenville palaeomagnetism and tectonics. Can. J. Earth Sci., 15: 687—695.

Morgan, G.E., 1976a. Palaeomagnetism of a slowly-cooled plutonic terrain in western Greenland. Nature, London, 259: 382—385.

Morgan, G.E., 1976b. The palaeomagnetism of some Precambrian rocks from the Itivdleq region of West Greenland. Ph.D. thesis, University of London, 356 pp.

Morgan, G.E. and Smith, P.K., 1981. Transmission electron microscope and rock magnetic investigations of remanence carriers in a Precambrian metadolerite. Earth Planet. Sci. Lett., 53: 226—240.

Nash, D., 1979. An interpretation of irregular dyke forms in the Itivdleq shear zone, West Greenland. Grøn. Geol. Unders., Rapp., 89: 71—84.

Noe Nygaard, A. and Ramberg, H., 1961. Geological reconnaissance map of the country between latitudes 69°N and 63°45′ N, West Greenland. Meddr. Grønnland, 124 (5), 9 pp.

Park, J.K., 1975. Palaeomagnetism of the Flin-Flon Snow Lake greenstone belt, Manitoba and Saskatchewan, Can. J. Earth Sci., 12: 1272—1291.

Park, J.K., 1977. A reconnaissance palaeomagnetic survey of the central Labrador Trough. Can. J. Earth Sci., 14: 159—174.

Piper, J.D.A., 1981. The altitude dependence of magnetic remanence in the slowly-cooled Precambrian plutonic terrain of West Greenland. Earth Planet Sci. Lett., 54: 449—466.

Pullaiah, G., Irving, E., Buchan, K.L. and Dunlop, D.J., 1975. Magnetisation changes caused by burial and uplift. Earth Planet. Sci. Lett., 28: 133—143.

Roy, J.L. and Blythe-Robertson, P., 1979. An assessment of the St. Urbain and other Grenville palaeopoles. Can. J. Earth Sci., 16: 1857—1865.

Schmidt, P.W., 1980. Palaeomagnetism of igneous rocks from the Belcher Islands, North West Territories, Canada. Can. J. Earth Sci., 17: 807—829.

Smith, R.L. and Piper, J.D.A., 1982. Palaeomagnetism of the Southern Zone of the Lewisian (Precambrian) Foreland, northwest Scotland. Geophys. J. R. Astron. Soc., 68: 325—347.

Stearn, J.E.F. and Piper, J.D.A., 1984. Palaeomagnetism of the Sveconorwegian belt, Fennoscandian Sield. Precambrian Res., 23: 201—246.

Ueno, H., Irving, E. and McNutt, R.H., 1975. Palaeomagnetism of the Whitestone anorthosite and diorite, the Grenville polar track and relative motions of the Laurentian and Baltic Shields. Can. J. Earth Sci., 12: 209—226.

Watterson, J.S., 1974. Investigations of the Nagssugtoqidian boundary in the Holsteinsborg district, central West Greenland. Grøn. Geol. Unders., Rapp., 65: 33—37.

Watson, J.V., 1973. Effects of reworking on high-grade gneiss complexes. Philos. Trans. R. Soc. London, Ser. A: 273: 443—455.